

RAFLY ANUGRAH SYAHPUTRA

Data Scientist | Machine Learning | Applied AI

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PROFESSIONAL SUMMARY

Data Science undergraduate with hands-on experience in machine learning, data engineering, and applied AI. Developed a CNN-LSTM rainfall nowcasting model at BMKG with a 0.88 F1-score and built operational data pipelines at PLN Pusdiklat. Independently built Autonomous AI Research Radar, AI Career Agent, StockPulse AI, and PersonaLab, spanning autonomous research, career intelligence, quantitative financial decision support, and evidence-grounded developer storytelling.

TECHNICAL SKILLS

Programming & Data: Python, SQL, R, JavaScript, Pandas, SQLAlchemy, PostgreSQL/pgvector, MySQL, SQLite

Machine Learning & AI: TensorFlow, Scikit-learn, NLP, CNN/LSTM, Statistical Modeling, Time Series, Gemini API, LangGraph, LangChain, RAG, LLM Agents

Data Engineering & Automation: ETL/ELT, Apache Airflow, Celery/RabbitMQ, Asyncio, Data Validation, Playwright, Selenium, Web Scraping

Analytics & MLOps: Power BI, Tableau, Looker Studio, Streamlit, Backtesting, Docker, Linux/systemd, Git/GitHub Actions, Pytest

RELEVANT EXPERIENCE

Data & Automation Intern | PT PLN (Persero), Pusdiklat

Apr 2026 - Aug 2026

- Built an ETL workflow that synchronized operational KPI data, processed 50+ Excel workbooks from 23 training units with Pandas/openpyxl, validated records, and stored clean outputs in SQLite for reporting.
- Developed KonMan, a five-page Streamlit dashboard for unit performance monitoring, data-quality checks, and automated PDF/Excel reporting.
- Created Looker Studio and Google Apps Script automations, plus a Selenium proof of concept for extracting Nota Dinas PDF data into outgoing-letter web forms; prepared Docker, systemd, and Synology handover guides.

Data Science Intern | BMKG, Juanda Meteorological Station

Jun 2025 - Dec 2025

- Automated a 24/7 Python ingestion workflow that replaced manual entry every 30 minutes and reduced operational workload by an estimated 80+ hours per month.
- Developed a CNN-LSTM model for 60-minute rainfall nowcasting using AWS observations and Himawari-9 satellite imagery, reaching 95% accuracy, 0.88 F1-score on significant weather events, and CSI 0.790 while outperforming baseline GRU models.
- Deployed real-time model outputs through an interactive Streamlit dashboard for operational review by forecasters.

SELECTED DATA SCIENCE & AI PROJECTS

StockPulse AI | Financial Intelligence & Quantitative Decision-Support System | [Live Showcase](#)

Python, Gemini API, Pydantic, SQLite, JavaScript, REST API, Systemd

- Architected a hybrid AI and deterministic decision pipeline for the Indonesia Stock Exchange, combining LLM-based news fact extraction with quantitative rules to reduce hallucinated financial reasoning.
- Engineered multi-source ingestion from 8+ tier-classified sources, official IDX foreign-flow data, and broker summaries with heuristic deduplication and date-conflict quarantine.
- Built an immutable prediction ledger and 5-layer evaluation framework for grounding, decision consistency, and H+5/H+20 forward outcomes with zero lookahead bias, realistic fees/slippage, and ATR-based risk levels.

Autonomous AI Research Radar | Evidence-First Autonomous Research Agent | [Github](#)

Python 3.12, Gemini, Hermes Agent Runtime, Pydantic v2, SQLite, Asyncio, Tenacity, Systemd, Pytest

- Architected an evidence-first research agent that separates Telegram UX, orchestration, and tool execution through authenticated HTTP boundaries, improving traceability and grounding.
- Built multi-source retrieval across arXiv, GitHub, Brave Search, and RSS with recency/velocity scoring, context-budgeted chunking, and SQLite deduplication.
- Implemented a copy-resistant consensus engine and a 32-case adversarial benchmark across 7 AI domains, measuring 100% grounding rate, citation coverage, and contradiction detection.

AI Career Agent | Human-in-the-Loop Career Intelligence System | [Live Showcase](#)

Python, LangGraph, LangChain, Gemini, Playwright, SQLAlchemy, SQLite, Pytest

- Architected a stateful career research workflow for multi-platform job discovery, vacancy scoring, and candidate-job matching using LangGraph and Gemini.
- Designed a grounded form-filling engine that evaluates dynamic application fields only against verified CV/profile facts, with human checkpoints for salary, visa, CAPTCHA, and other sensitive inputs.
- Implemented multi-tier Gemini routing, automatic rate-limit failover, 30-day response caching, and persistent workflow state to improve reliability and control API usage.

PersonaLab | AI-Powered Developer Storytelling & Branding Platform

Next.js, TypeScript, Gemini API, GitHub API, LinkedIn REST API, Server Actions

- Built an end-to-end platform that transforms GitHub commits, code diffs, and repository READMEs into factual, publication-ready LinkedIn updates.
- Implemented a 4-stage AI pipeline (Content Scout, Evidence Judge, Draft Writer, Final Reviewer) and cross-repository fact checks to prevent fabricated metrics or business-impact claims.
- Added repository-specific project memory and deterministic commit-SHA hashing to avoid repetitive narratives and redundant LLM calls, with GitHub extraction and LinkedIn publishing integration.

EDUCATION & RECOGNITION

B.Sc. Data Science | State University of Surabaya (UNESA)

Aug 2023 - Present

GPA 3.67/4.00 | 122 credits

- Finalist & Co-Author, SENADA 2025 National Seminar - distributed ETL pipeline for clickbait and news classification using Celery and RabbitMQ.
- Semifinalist, WordPress Case Competition 2024 - developed a data-driven digital talent strategy for Society 5.0.
- Finalist, FMIPA Ambassador Selection 2024-2025; awarded the title of Putra Konservasi.

LEADERSHIP & ACTIVITIES

Commission A Staff | Student Representative Council (DPM) FMIPA UNESA

Feb 2025 - Jan 2026

- Monitored student-executive work programs, evaluated organizational performance, and served as MC/moderator for faculty forums and seminars.

Official Faculty Ambassador | FMIPA Ambassador Association

Sep 2024 - Sep 2025

- Represented the faculty at university events and coordinated cross-faculty communication, event logistics, and community initiatives.

LANGUAGES

Indonesian (Native) | English (Intermediate)